

Spring Data JPA – Quick Example

Spring Data JPA is a part of the Spring Framework that simplifies database access by reducing the amount of boilerplate code required for data persistence.

1. Create an Entity Class

An entity class represents a table in the database.

```
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
@Entity
public class Book {
    @Id
    private int id;
    private String title;
    public Book() {}
    public Book(int id, String title) {
        this.id = id;
        this.title = title;
    }
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = id;
    }
    public String getTitle() {
        return title;
    }
    public void setTitle(String title) {
        this.title = title;
    }
}
```

2. Create a Repository Interface

The repository interface extends `JpaRepository` and provides built-in CRUD methods without requiring any implementation.

```
import org.springframework.data.jpa.repository.JpaRepository;
public interface BookRepository extends JpaRepository<Book, Integer> {
}
```

3. Perform CRUD Operations

@Autowired

```
private BookRepository bookRepository;
```

```
bookRepository.save(new Book(1, "Spring Framework"));
```

```
bookRepository.findAll();
```

```
bookRepository.deleteById(1);
```

Difference between JPA, Hibernate, and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
Definition	A Java specification for Object Relational Mapping (ORM).	An ORM framework that implements JPA.	A Spring module that simplifies data access using JPA.
Type	Specification	Framework	Spring Framework Module
Developed By	Oracle/Java Community	Hibernate Team	Spring Team
Implementation	Does not provide implementation.	Provides complete implementation of JPA.	Uses JPA implementation (such as Hibernate) internally.
SQL Generation	No	Yes	Yes
CRUD Operations	Not directly available	Available through Session API	Available automatically through JpaRepository